(6) $\dfrac{21+7\sqrt{2}+3\sqrt{6}+2\sqrt{3}}{7}$ (7) 5 (8) $\dfrac{2+2\sqrt{6}}{5}$

Exercise 2.12

(1) $\log_b y = x$, $(0,\infty)$, $(-\infty,\infty)$

(2) $\dfrac{5}{6}$ (3) 64 (4) 2 (11) $2\sqrt{2}$ (12) -10

Exercise 2.13

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(2)	(1)	(2)	(3)	(2)	(2)	(3)	(1)	(2)	(2)	(2)	(3)	(3)	(2)	(3)

16	17	18	19	20
(1)	(3)	(1)	(1)	(4)

Exercise 3.1

(1)(i) first quadrant (ii) second quadrant (iii) fourth quadrant (iv) fourth quadrant (v) second quadrant

(2)(i) 35° (ii) 165° (iii) 70° (iv) 90° (v) 270° (8) $k \in [-1,1]$

(9) $\sec\theta = \dfrac{p^2+1}{2p}$; $\tan\theta = \dfrac{p^2-1}{2p}$; $\sin\theta = \dfrac{p^2-1}{p^2+1}$ (12) $\left(c^2+bd\right)^2 = (ad+cb)^2 + \left(ac-b^2\right)^2$

Exercise 3.2

(1)(i) $\dfrac{\pi}{6}$ radians (ii) $\dfrac{3\pi}{4}$ radians (iii) $-\dfrac{41\pi}{36}$ radians (iv) $\dfrac{5\pi}{6}$ radians (v) $\dfrac{11\pi}{6}$ radians

(2)(i) 60° (ii) 20° (iii) 72° (iv) 420° (v) 200°

(3) $r \approx 31.82$ meters (4) $s = \dfrac{20\pi}{3} = 20.92$cm (5) $\theta = 12^\circ 36'$ (6) $s = 7.16$feet

(7) $r_1 : r_2 = 5 : 4$

(8) Angle of sector $\approx 65^\circ 27' 16''$ (9) 6000° (10) 14° (11) $\dfrac{3\pi}{4}$

Exercise 3.3

(1)(i) $\dfrac{\sqrt{3}}{2}$ (ii) $-\dfrac{1}{2}$ (iii) $\dfrac{1}{2}$ (iv) $-\dfrac{1}{\sqrt{3}}$ (v) $-\dfrac{1}{\sqrt{3}}$ (vi) $\sqrt{3}$ (vii) $\dfrac{\sqrt{3}}{2}$

(2) $\sin\theta = \dfrac{2\sqrt{6}}{7}$; $\cos\theta = \dfrac{5}{7}$; $\tan\theta = \dfrac{2\sqrt{6}}{5}$;

$\operatorname{cosec}\theta = \dfrac{7}{2\sqrt{6}}$; $\sec\theta = \dfrac{7}{5}$; $\cot\theta = \dfrac{5}{2\sqrt{6}}$